

Programming Assignment 1 (Deep Learning From Scratch)

In this assignment we implement a Deep Neural Network

Implement **from scratch** (hence not using NN libraries such as tensorflow, keras, pytorch) a Neural Network that is able to approximate a function of two variables $f(x, y)$. The network will have at least two hidden layers and it is trained using the backpropagation algorithm. The test case we consider is $f(x, y) = x \cdot y$ (we can say that we teach the network to multiply two numbers).

As training data for the algorithm construct a matrix of input output triplets of the function $f(x, y) = x \cdot y$. When $x, y \in [0, 1]$ the matrix would contain training triplets of the form $[x_i, y_j, x_i \cdot y_j]$, where x_i and y_j are random values in $[0, 1]$.

Try different activation functions and find which one performs best with your network. For validation accuracy calculate the root mean square error.

Add visualization of the RMSE over training epochs, and perform simple tests to check accuracy.